



Debiasing Movie Recommendation System using Edge Dropout

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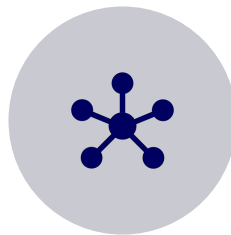
Agenda



PROBLEM
DEFINITION



RESEARCH
QUESTION



GRAPH NEURAL
NETWORK



DISCUSSION



PROBLEM DEFINITION

Importance of group-fairness in recommendation



Lack of social diversity



Job opportunities



Filter Bubble effect

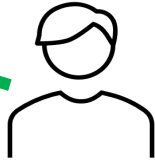
Motivation



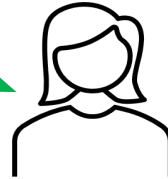
[1.]



[3.]



[3.]



[2.]



[4.]

Research Question

RQ1. Do diff GNN architectures effect Fairness differently?

RQ2. How can we remove gender gender bias in Movie Recommendation Systems trained using GNNs?

Dataset Preparation – MovieLens 100k

	userId	movieId	rating	timestamp
5	298	474	4	884182806
7	253	465	5	891628467
11	286	1014	5	879781125
12	200	222	5	876042340
16	122	387	5	879270459

User id | Movie id | Rating | Timestamp

	userId	age	gender	occupation	zip
0	1	24	M	technician	85711
1	2	53	F	other	94043
2	3	23	M	writer	32067
3	4	24	M	technician	43537
4	5	33	F	other	15213

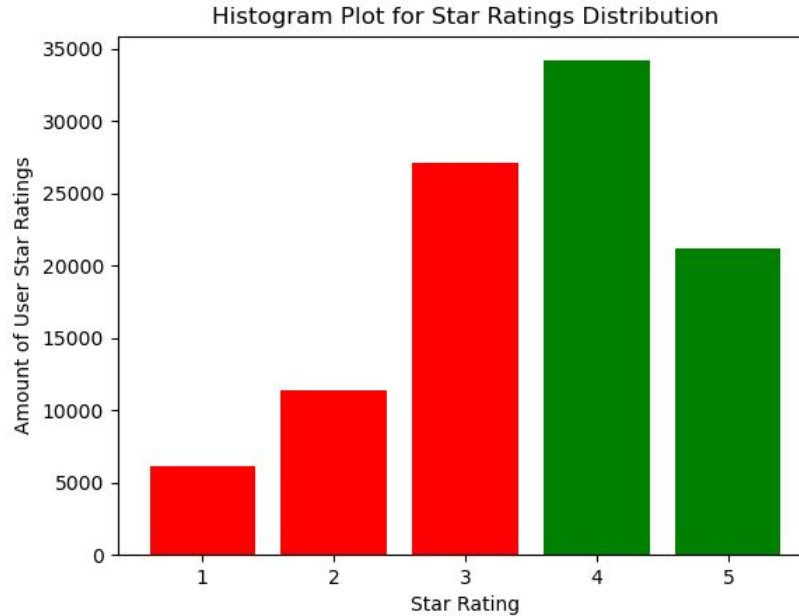
User id | Age | Gender | Occupation | Zip
code

Dataset Preparation – MovieLens 100k

	title	video_release_date	IMDb_URL	unknown	Action	Adventure	Animation	Children	Comedy	Crime	...	Fantasy
movieId												
1	Toy Story (1995)	01-Jan-1995	http://us.imdb.com/M/title-exact?Toy%20Story%20(1995)	0	0	0	1	1	1	0	...	0
2	GoldenEye (1995)	01-Jan-1995	http://us.imdb.com/M/title-exact?GoldenEye%20(1995)	0	1	1	0	0	0	0	...	0
3	Four Rooms (1995)	01-Jan-1995	http://us.imdb.com/M/title-exact?Four%20Rooms%20(1995)	0	0	0	0	0	0	0	...	0
4	Get Shorty (1995)	01-Jan-1995	http://us.imdb.com/M/title-exact?Get%20Shorty%20(1995)	0	1	0	0	0	1	0	...	0
5	Copycat (1995)	01-Jan-1995	http://us.imdb.com/M/title-exact?Copycat%20(1995)	0	0	0	0	0	0	1	...	0

Movie id | Name | Release Date | imdb link | Genres

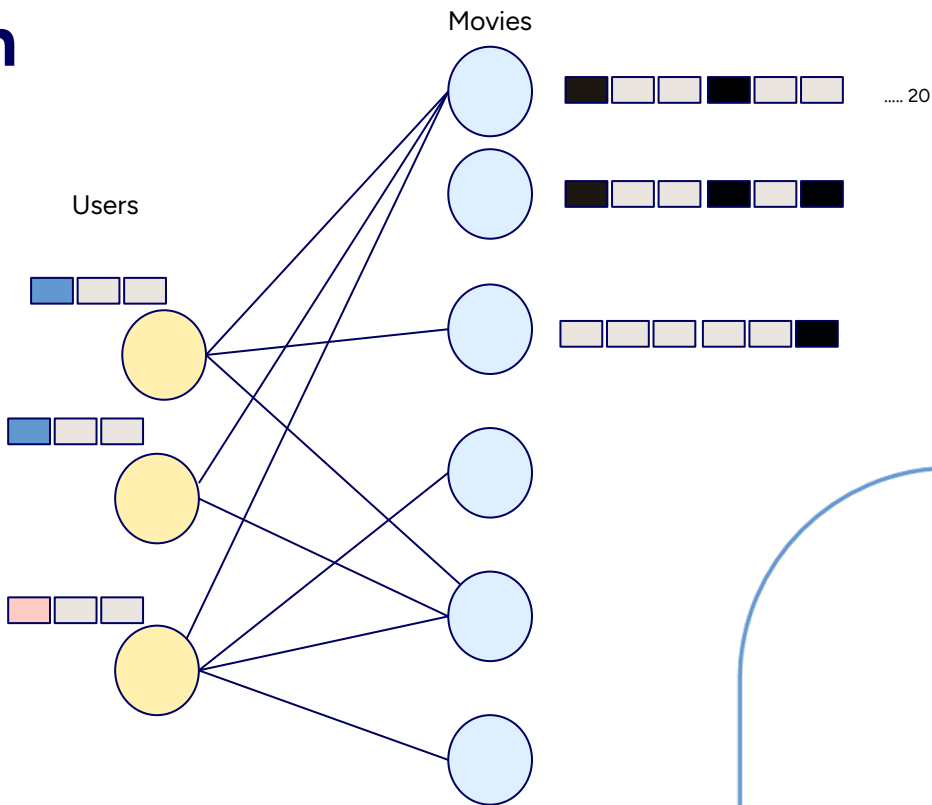
Dataset Preparation

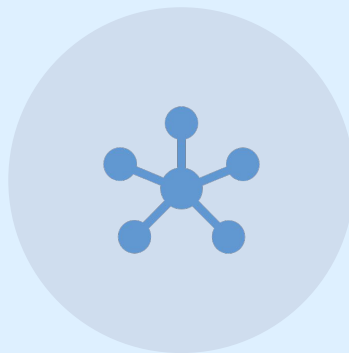


Dataset Preparation

M Occupation 23

F





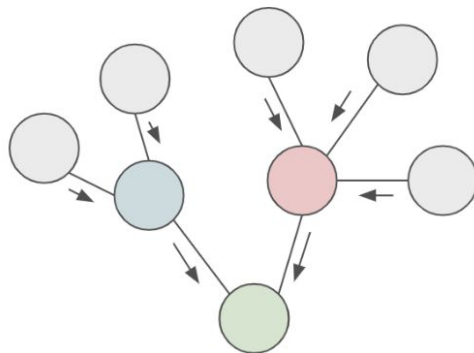
GRAPH NEURAL NETWORK

How GNNs work?

- **GNNs** learn by aggregating information from their neighbouring nodes through an algorithm called **Message Passing** (which we won't go into detail).
- An normal Deep Neural Network does not take into account structural properties in their representations directly.

GCN

$$x'_i = \mathbf{W}_1 x_i + \mathbf{W}_2 \sum_{j \in \mathcal{N}(i)} e_{j,i} \cdot x_j$$

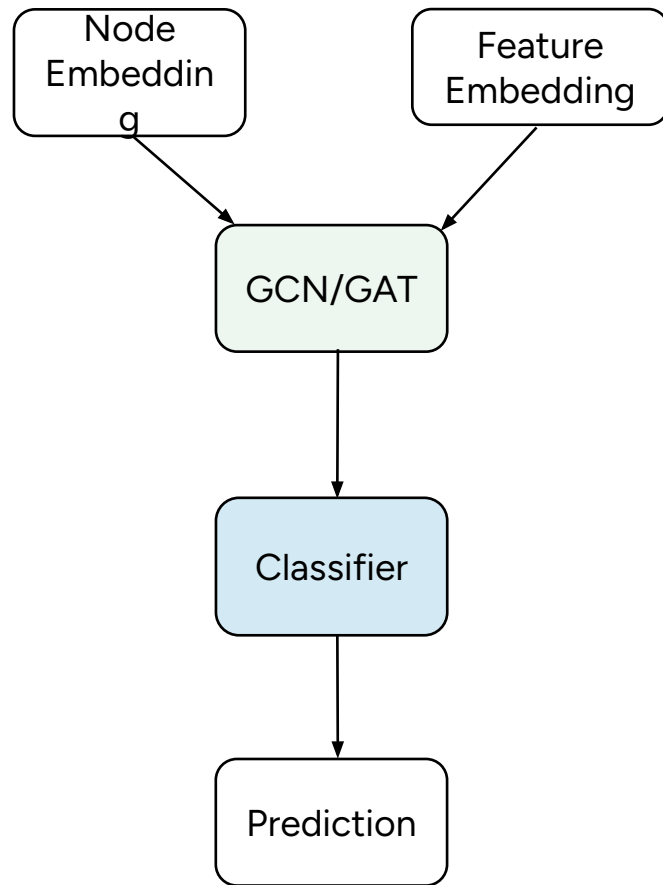


GAT

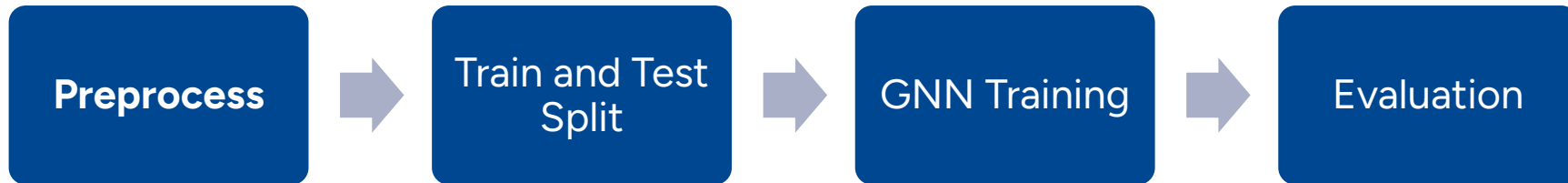
$$\mathbf{x}'_i = \alpha_{i,i} \Theta_s \mathbf{x}_i + \sum_{j \in \mathcal{N}(i)} \alpha_{i,j} \Theta_t \mathbf{x}_j$$

GNN model development

PyTorch Geometric was used as a library in Python for developing and coding GNNs.



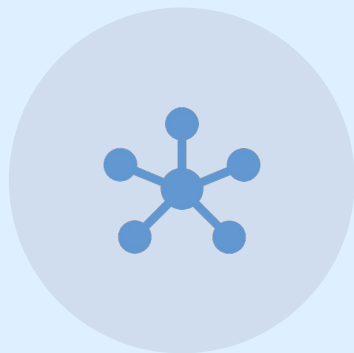
Model Training



Why we simply cannot remove Gender attributes

- Removing Gender attributes significantly reduces accuracy.

Feature Ablation	Accuracy	F1	AUC	MAP	Statistical Parity	Equality of Opportunity
Age + Occ + Gender–Movie	0.7998	0.6809	0.8643	0.5458	0.036829	0.047949
Age + Occ– Movie	0.7971	0.6837	0.8635	0.5572	0.026296	0.030179
Age–Movie	0.7310	0.4290	0.8061	0.2591	0.020913	0.021649

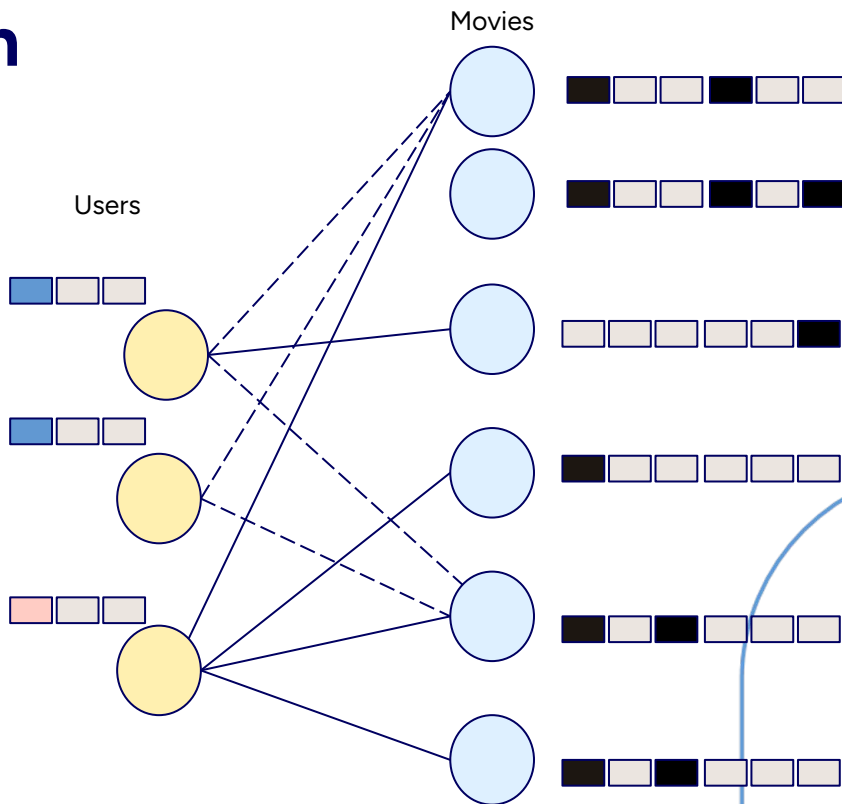


Adapting FairDrop

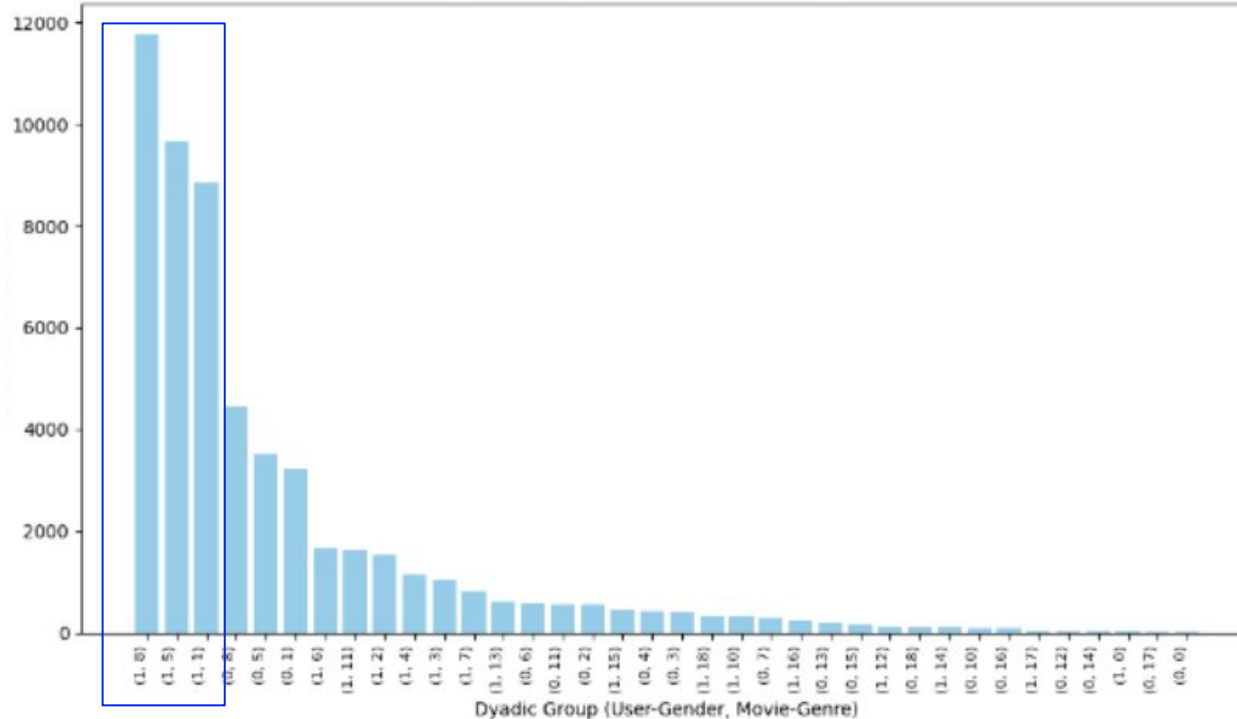
FairDrop adaptation

M Occupation 23

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Dyadic Grouping of Edges for Fair Edge Drop



Drop Edges randomly **BUT**
with **higher probability** from
groups which cause bias:
Male - Action
Male - Adventure
Male - SciFi

Dyadic Fairness metrics

- **Statistical Parity**

- Assesses whether the proportion of movie-recommendation is the same for all groups
- Formula:

$$|P(Y_{\hat{}} = 1|G=\text{Female}) - P(Y_{\hat{}} = 1|G=\text{Male})|$$

- **Equality of Opportunity**

- Assesses whether the probability of a movie-recommendation is the same for individuals from different groups who have the same qualifications.
- Formula:

$$|P(Y_{\hat{}} = 1|Y=1,G=\text{Female}) - P(Y_{\hat{}} = 1|Y=1,G=\text{Male})|$$

RQ1: Results – Effect of GNN Architectures

GAT: Dynamic Arch → Fairness increasing with more layers ; Accuracy decreases with more layers.

GCN: Fixed Arch → Fairness decreases with more layers. ; Accuracy increases with more layers.

TABLE III: Effect of layer-depth on GAT & GCN architectures.

Layers	Loss	Acc ↑	AUC ↑	F1 ↑	Pr	Rec	MAP	$\Delta SP_s \downarrow$	$\Delta EO_s \downarrow$	$RB_{Node} \downarrow$
GAT _{L2}	0.3840	0.7952	0.8727	0.6911	0.6874	0.6949	0.5783	0.7692 (0.3170)	1.0000 (0.422, 0.1351)	0.487
GAT _{L4}	0.4489	0.7894	0.8674	0.6617	0.6180	0.7120	0.5234	0.6000 (0.2616)	0.8571 (0.5415, 0.1016)	-
GCN _{L2}	0.4522	0.7609	0.8296	0.6589	0.6928	0.6282	0.5719	0.7846 (0.3523)	1.0000 (0.6322, 0.2044)	0.52
GCN _{L4}	0.4664	0.8006	0.9143	0.7415	0.8582	0.6527	0.7124	0.9846 (0.3058)	1.0000 (0.8065, 0.2530)	0.4887

GCN: Fixed Arch → Fairness decreases with more layers. ; Accuracy increases with more layers.

HOPS

RQ2:Results – Random Edge Drop vs FairDrop

TABLE V: Performance measures for Random Edge Drop (RD) and FairDrop (FD) on GAT and GCN.

Method	Loss	Acc $\uparrow$	AUC $\uparrow$	F1 $\uparrow$	Pr	Rec	MAP	$\Delta SP_s \downarrow$	$\Delta EO_s \downarrow$	$RB_{Node} \downarrow$
GAT + RD	0.3873	0.7979	0.8689	0.6960	0.6941	0.6980	0.5846	0.9831 (0.3060)	0.8824 (0.5944, 0.1311)	0.5056
GAT + FD _{0.55}	0.3878	0.7896	0.8672	0.6578	0.6066	0.7184	0.5151	0.9796 (0.3019)	1.0000 (0.6275, 0.1777)	0.5147
GAT + FD _{0.6}	0.4053	0.7694	0.8548	0.6079	0.5364	0.7015	0.4523	0.5000 (0.2334)	0.8750 (0.4830, 0.0989)	0.483
GAT + FD _{0.65}	0.3997	0.7743	0.8548	0.6214	0.5556	0.7049	0.4692	0.9583 (0.3058)	0.8333 (0.5851, 0.1311)	0.5105
GAT + FD _{0.7}	0.4014	0.7776	0.8540	0.6489	0.6168	0.6846	0.5170	0.4831 (0.2941)	1.0000 (0.6145, 0.1448)	0.496
GCN + RD	0.4728	0.7074	0.7803	0.4934	0.4274	0.5835	0.3511	0.4787 (0.2414)	0.6667 (0.3906, 0.1415)	0.52
GCN + FD _{0.55}	0.4529	0.7451	0.8136	0.6605	0.7438	0.59	0.6113	0.9310 (0.4055)	1.0000 (0.7263, 0.3185)	0.4957
GCN + FD _{0.6}	0.4646	0.7633	0.8317	0.6672	0.7118	0.6278	0.5875	0.8571 (0.3661)	0.8750 (0.6634, 0.3165)	0.477
GCN + FD _{0.65}	0.4646	0.7592	0.8265	0.6356	0.6299	0.6414	0.5214	0.7808 (0.3232)	0.8571 (0.5697, 0.2946)	0.5
GCN + FD _{0.7}	0.3617	0.8172	0.8847	0.7065	0.6600	0.7601	0.5711	0.5000 (0.2727)	1.0000 (0.6016, 0.1177)	0.507



Critical Reflection

Critical Reflection & Implications of Debiasing

- Movie Datasets contain biases due to combination of **over-representation of males** but also due to **Popularity** of certain genres over the another.
- Current **Fairness metrics** require improvement as optimizing our models for them
- Adapted FairDrop method might lead to loss of personalization , which might in turn lead
- Fairness objective functions might remove gender effect altogether.
- Reduction in Fairness is usually going to reduce Accuracy slightly ~ **2/5%**

Future Work

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- This might or might-not be desirable.

References

1. F. Masrour, T. Wilson, H. Yan, P.-N. Tan, and A. Esfahanian, "Bursting the filter bubble: Fairness-aware network link prediction," Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, pp. 841–848, Apr. 2020.
2. [Our Model Achieves Excellent Performance on MovieLens: What Does It Mean? \(arxiv.org\)](#)
3. I. Spinelli, S. Scardapane, A. Hussain and A. Uncini, "FairDrop: Biased Edge Dropout for Enhancing Fairness in Graph Representation Learning," in IEEE Transactions on Artificial Intelligence, vol. 3, no. 3, pp. 344-354, June 2022, doi: 10.1109/TAI.2021.3133818